



**Ningyi Li**

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## EDUCATION

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- **Xiamen University, School of Mathematical Sciences** 09/2020 - 06/2024  
*Bachelor of Science / Mathematics and Applied Mathematics (major)* Xiamen, China
  - Average score: 83.34/100
  - Advisor: Prof. Wenfei Liu
  - Graduate-level Courses: Commutative Algebra, Homological Algebra, Differential Manifold, Algebraic Graph Theory
- **Xiamen University, The Wang Yanan Institute for studies in Economics (WISE)** 09/2020 - 06/2024  
*Bachelor of Economics / Mathematical Finance (minor)* Xiamen, China
  - Average score: 86.15/100
  - Advisor: Prof. Linlin Niu
- **University of Warsaw, Faculty of Mathematics, Informatics and Mechanics** 02/2023 - 07/2023  
*Exchange Student / Mathematics* Warsaw, Poland
  - Average score: 4.00/5.00
  - Graduate-level Courses: Algebraic Geometry, Algebraic Topology, Lie Groups and Lie Algebras, Functional Analysis
- **Universiteit Leiden (ALGANT Master Program)** 2024 - 2025  
*Master of Science / Algebra, Geometry and Number Theory* Leiden, Netherlands
  - Expected.
- **Universität Regensburg (ALGANT Master Program)** 2025 - 2026  
*Master of Science / Algebra, Geometry and Number Theory* Regensburg, Germany
  - Expected.

## ACADEMIC EXPERIENCES

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- **Tianyuan Mathematical Center in Southeast China, Summer School**  
(jointly held by Xiamen University, Zhejiang University and Sun Yat-sen University) 07/2023  
*Student / Algebra, Geometry and Analysis* Xiamen, China
  - Average score: 100/100
  - Courses: Galois Theory, An Introduction to Atiyah-Singer Index Theorem, Liouville/Bernstein theorems for Laplace/minimal surface, maximal surface and Monge-Ampere equations
- **Shanghai Center for Mathematical Sciences, IWoAT Summer School 2024** 06/2024  
*Student / Algebraic Topology* Shanghai, China
  - Courses: Stable Motivic Homotopy Theory
- **University of Science and Technology of China, Summer School in Geometry** 07/2024  
*Student / Geometry* Hefei, China
  - Courses: Geometry and dynamics of Arnold's cat map, Some aspects of Ricci flow on non-compact manifolds, Topics in mean curvature flow, An overview of the barycenter method

RESEARCH EXPERIENCES

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| <b>•Undergraduate Thesis (major)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 05/2024                |
| Supervised by Prof. Wenfei Liu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Xiamen, China          |
| <i>Freudenthal Compactification and Its Application in Mapping Class Groups of Noncompact Surfaces.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        |
| <ul style="list-style-type: none"><li>– Presented the extension theorems for proper continuous maps, homeomorphisms, and isotopies on Freudenthal compact spaces.</li><li>– Established a one-to-one correspondence between homeomorphisms and isotopies on the original space and its Freudenthal compactification.</li><li>– Showed the one-to-one correspondence between the punctures of a surface and its ends, thereby obtaining that the mapping class group of a punctured surface is isomorphic to the group of isotopy classes of orientation-preserving homeomorphisms of its Freudenthal compactified surface that keep the set of ends fixed.</li><li>– As a simple application of this conclusion, I computed the mapping class groups of the sphere and the sphere with one, two and three punctures, respectively.</li></ul> |                        |
| <b>•Undergraduate Thesis (minor)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 04/2024                |
| Supervised by Prof. Linlin Niu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Xiamen, China          |
| <i>Effects of Daily Exercise Time on the Academic Performance of Students: An Empirical Analysis Based on CEPS Data.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |
| <ul style="list-style-type: none"><li>– By dividing the daily exercise time into five sections to construct a categorical variable in a linear regression model as well as using another model to draw intuitive figures, I find that spending too little or too much time on physical activity every day would have adverse impacts on students academic performance, with differences existing in the impacts by gender, grade, city scale, and location type of the school.</li></ul>                                                                                                                                                                                                                                                                                                                                                     |                        |
| <b>•Short Essay</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 11/2024                |
| Joint with Gabriele Reali   Advised by Prof. Yajnaseni Dutta and Dr. Emma Brakkee                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Amsterdam, Netherlands |
| <i>Notes on Serre Duality.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                        |
| <ul style="list-style-type: none"><li>– In this short notes, we present the the Serre duality (for Hom and for Ext) for projective spaces, and then extend the result for smooth projective varieties.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                        |

NONACADEMIC ACTIVITIES

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| <b>•Volunteer</b> , Centennial Anniversary of Xiamen University | 2021 |
| <b>•Member</b> , Xiamen Marathon                                | 2021 |

AWARDS AND HONORS

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|-----------------------------------------------|------|
| <b>•Talent Training Plan Scholarship</b>      | 2021 |
| <b>•Undergraduate Outstanding Scholarship</b> | 2021 |
| <b>•Undergraduate Outstanding Scholarship</b> | 2023 |

SKILLS

**Languages:** Chinese (Native), English (IELTS: 7)  
**IT:** MATLAB, Mathematica, Stata  
**Tools:** Latex